Weekly Report: Ultra-High-Pressure ODMR — Is the 475 nm optimal blue wavelength robust to detection-window background?

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Week: 2026/07/26 – 2026/08/01

Question

Our NV-only model puts the optimal blue excitation wavelength at 120 GPa at $\lambda_{opt} \approx 475$ nm; does adding a spin-independent background inside the > 650 nm detection window shift λ_{opt} far enough to change which commercial laser line (473 or 488 nm) we should order?

Hypothesis

If the Cr^{3+} transmission gap (∼476 nm) sits close to the NV-only optimum while N3 background rises toward the near-UV, background should push λ_{opt} only slightly to the red, staying inside the existing 5% sensitivity tolerance band so that the choice between 473 nm and 488 nm laser lines is unaffected.

Evidence / Interpretation

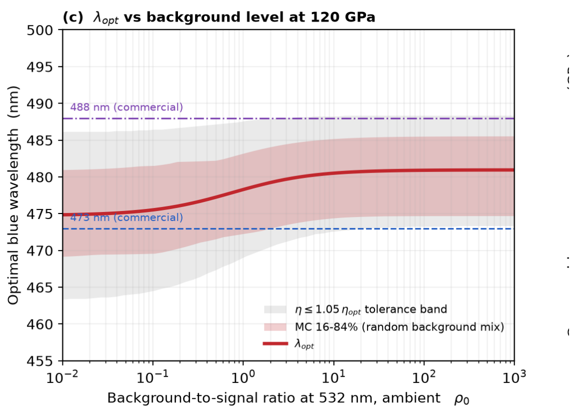

図 1: Optimal blue wavelength λ_{opt} vs. background level ρ_0 at 120 GPa, with the $\eta \leq 1.05\,\eta_{opt}$ tolerance band (grey) and a Monte-Carlo 16–84% band over random background compositions (red shading).

- λ_{opt} shifts only from 474.8 nm (ρ_0=0) to 480.9 nm ($\rho_0 \to \infty$), i.e. < 6 nm total, and stays inside the zero-background tolerance band [463, 486] nm.

- The Monte-Carlo band over randomized ruby/N3/broad-luminescence mixing weights spans roughly [465, 487] nm across all ρ_0, so the conclusion does not depend on which background composition we assume.

Gap / Failure

- Both 473 nm and 488 nm remain within the tolerance band for every ρ_0 tested, so this calculation alone cannot rank the two commercial lines against each other.
- The model treats background as wavelength-shape only and ignores any pressure dependence of the ruby/N3 emission strength itself, which could move the gap location at 120 GPa.

Next Step

Once the background spectrum is measured (see companion report on the crossover question), refit the Cr^{3+}/N3 shape parameters to the real anvil and re-run `fig5_background.py` to check whether the calibrated λ_{opt} still favors 473 nm over 488 nm.

References

1. K. O. Ho, C. Dailledouze, V. Žalandauskas, *et al.*, "Optical Stability and Photophysics of NV Centers in Diamond up to 120 GPa," arXiv:2606.02399 (2026).
2. A. Dréau *et al.*, Phys. Rev. B **84**, 195204 (2011).
3. Repository: `code/nv_bg.py`, `code/background.py`, `code/fig5_background.py` (this analysis).